

Supplement: Lineage Tables for the 10,240-Byte Generator

The DOPE Project

Abstract—Lineage-level retention and byte charges for the displayed profile, the 8,192-step sufficiency budget, and the 142 BeyondArena family names. These tables are the rows behind the medians in the main paper. A blank retention is an unavailable fit or an auditor that is not informative on all three sample seeds. It is not a zero. BeyondArena is not pooled with the PMLB panel. Master Fitness v2 is not in these tables.

APPENDIX A

DISPLAYED PROFILE, ALL 100 LINEAGES

TABLE I: Displayed profile, all 100 lineages.

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
02a45777900441ce	215_2dplanes	800	10	2,583	pass	0.9930	1.0001	0.9990	1.0477
047e62b52a3f4ede	695_chatfield_4	156	12	3,311	pass	1.0041	1.0155	1.0048	—
04c80944e971cb3e	first_principles_supernovae_zr	157	1	762	pass	0.8753	0.8846	0.9959	1.2627
0561d673cb73c27e	666_rmftsa_ladata	339	10	2,921	pass	1.0095	1.1026	1.0338	1.1394
060b9fff8092db17	589_fri_c2_1000_25	667	25	5,717	pass	0.8893	0.9534	0.9911	1.5794
06b0312a585b436e	1196_BNG_pharynx	800	10	2,631	pass	0.8601	0.8964	1.0458	1.7049
0779d3fd5b3133a2	nikuradse_2	241	1	758	pass	0.9096	0.9261	2.1723	2.2082
07f38946316c039b	593_fri_c1_1000_10	667	10	2,876	pass	0.5650	0.6068	0.9617	1.6005
0a8428ef25a69175	537_houses	800	8	2,609	pass	0.6257	0.7036	1.0150	1.0256
0d482c1dd7cc6592	210_cloud	72	5	1,844	pass	1.0374	0.9772	1.0944	1.0239
0db59fd2579600d0	665_sleuth_case2002	98	6	1,962	pass	1.0371	1.1324	1.1345	—
0f86a714ec5f3ac6	1191_BNG_pbc	800	18	4,183	pass	0.8097	0.8761	0.9173	—
167578f5deba4977	547_no2	333	7	2,443	pass	0.8073	0.8513	1.0097	0.9898
19f2c3c2d67defd7	feynman_II_11_27	800	4	1,494	pass	0.9704	0.9890	0.9960	1.0048
19f4780b53b3fa41	678_visualizing_environmental	74	3	1,515	pass	—	—	—	0.4431
1a4456d1cc71a26a	feynman_test_1	800	7	2,191	pass	1.0608	1.1671	0.9816	—
1a803b27c2400dc4	feynman_III_10_19	800	4	1,520	pass	0.9882	0.9952	0.9942	0.9965
1cb97b1d7314d43d	229_pwLinear	133	10	2,587	pass	0.8417	0.9878	0.9104	—
205f4b4982b9ea49	608_fri_c3_1000_10	667	10	2,845	pass	0.5263	0.5671	1.0451	0.8182
26eff843be35d831	581_fri_c3_500_25	333	25	5,647	pass	0.7602	0.9020	1.1297	1.5792
282f8ddb43b5e677	595_fri_c0_1000_10	667	10	2,904	pass	0.8974	0.9234	0.9962	1.0250
297b7688a045c583	603_fri_c0_250_50	167	50	10,171	pass	0.8848	1.1562	1.0700	2.6883
2c8d0985f3557d2b	feynman_II_11_28	800	2	1,026	pass	0.9831	0.9899	1.0031	0.9976
30ca2887550f0d8d	201_pol	800	26	8,145	pass	0.5742	0.5976	0.7015	0.6923
3408a6a513c627a3	strogatz_barmag1	267	2	1,011	pass	0.9750	0.9775	0.9969	1.0058
35a9aa9b88f0e4d1	344_mv	800	10	2,737	pass	0.9985	1.0016	0.9662	1.0347
3d3321e27d14f9a4	599_fri_c2_1000_5	667	5	1,707	pass	0.8359	0.8611	0.9668	1.6797
44eaabe4cb923766	613_fri_c3_250_5	167	5	1,712	pass	0.8671	0.9276	0.8599	—
46b25db9396c8209	227_cpu_small	800	12	3,327	pass	0.8569	0.9072	0.9018	0.9364
46c5a9e5ae42037	1193_BNG_lowbw	800	9	2,436	pass	0.9635	0.9703	1.0030	0.9871
471f11cc2d38853a	feynman_test_11	800	4	1,481	pass	0.9645	0.9760	0.9975	0.9826
47c581af5de1136b	582_fri_c1_500_25	333	25	5,670	pass	0.7510	0.9088	0.7173	—
48b7a3bf81e5f3e5	605_fri_c2_250_25	167	25	5,692	pass	0.4046	0.6465	1.9138	—
4a354d98c965b01f	597_fri_c2_500_5	333	5	1,713	pass	0.7550	0.8055	0.9814	1.8381
4aa84fff8ad0cd26	feynman_III_13_18	800	4	1,501	pass	0.8748	0.9137	0.8850	0.9348
51730b348b1bbd14	557_analcatdata_apnea1	316	3	1,237	pass	-0.0032	0.0086	—	—
54b221668673164c	feynman_III_14_14	800	5	1,726	pass	1.0346	1.1164	1.0011	1.0611
552665868f8f72ed	strogatz_vdp2	267	2	989	pass	0.9945	0.9963	0.9960	0.9969
585fef2a1cb6b33f	strogatz_lv2	267	2	1,018	pass	0.8286	0.9859	0.7502	—
642738a361f1c63a	feynman_I_12_1	800	2	1,032	pass	0.9892	0.9910	0.9892	0.9933
64c64f0f9983c48f	strogatz_bacres2	267	2	998	pass	0.9793	0.9832	0.9783	0.9862
654566a1f91408aa	4544_GeographicalOriginalofMusic	706	117	22,233	fail	—	—	—	—
693890aae9dc860b	first_principles_supernovae_zg	162	1	761	pass	0.8492	0.8908	1.0066	1.1997
6cacb11611345a24	feynman_I_11_19	800	6	1,947	pass	0.9524	0.9812	0.9780	0.9800
7291b347fe0238fb	feynman_II_11_20	800	5	1,757	pass	0.9928	1.0432	1.0016	1.0674
7423783f7e38bb9b	strogatz_shearflow1	267	2	1,021	pass	0.4900	0.5614	-0.3978	—
7a3d123dd555110d	598_fri_c0_1000_25	667	25	5,668	pass	0.8613	0.9166	1.0046	1.0034
7c8943b160464add	601_fri_c1_250_5	167	5	1,719	pass	0.7859	0.8552	1.0773	—
7e51711baa1b1fb1	feynman_I_12_11	800	5	1,718	pass	0.9267	0.9320	1.0015	0.9860
80589368fa78f21a	218_house_8L	800	8	2,392	pass	0.5835	0.6708	0.9616	1.1040
8164950b155f7f56	feynman_I_12_2	800	4	1,511	pass	0.9702	0.9986	0.9485	0.9753
85a31ae409944a1b	feynman_II_10_9	800	3	1,269	pass	0.9330	0.9508	0.9589	0.9515
8d859cfa7b51a351	feynman_III_15_12	800	3	1,248	pass	0.5172	0.5542	1.0228	1.0749
923483635406b213	1203_BNG_pwLinear	800	10	2,581	pass	1.2079	1.1750	0.9941	1.5230
9312d813229df035	690_visualizing_galaxy	216	4	1,458	pass	0.9725	0.9755	0.9853	1.0870
94a00ea67f49bf84	596_fri_c2_250_5	167	5	1,698	pass	0.7546	0.8072	—	—
94bd4520b8ceb3f0	feynman_test_12	800	5	1,747	pass	0.9688	0.9804	0.9913	1.0067
956d34f13e7af830	586_fri_c3_1000_25	667	25	5,701	pass	0.9325	0.9849	1.2339	2.1457
9a10c15885022034	_deprecated_195_auto_price	106	15	5,025	pass	0.8717	0.9485	0.9598	1.3107
a04f964bc53281e0	strogatz_lv1	267	2	985	pass	-3.8784	-3.4212	—	—
a23f4bd626b76590	feynman_test_13	800	5	1,740	pass	0.9178	0.9605	0.9900	1.0991
a32a69e7c701eb14	588_fri_c4_1000_100	667	100	19,019	fail	—	—	—	—
a5ba682572f53add	602_fri_c3_250_10	167	10	2,900	pass	0.5355	0.8109	0.7988	—
a863cd42be4d05d4	663_rabe_266	80	2	1,014	pass	0.9781	0.9933	0.9967	1.0521
a97065df07a47eb0	592_fri_c4_1000_25	667	25	5,655	pass	0.8791	0.9340	0.9578	1.2714
ab279eded3b3c52b	574_house_16H	800	16	4,035	pass	0.1181	0.1796	0.1795	0.3705
aebf5d861a8365b4	522_pm10	333	7	2,409	pass	0.5082	0.6149	0.4125	2.3076
b6adf79118a58cea	562_cpu_small	800	12	3,353	pass	0.6968	0.7747	0.9834	0.9307
b78d323566930bd7	583_fri_c1_1000_50	667	50	10,098	pass	0.8607	0.9444	0.9960	1.6211
b7f9774e2142a758	strogatz_shearflow2	267	2	1,019	pass	0.4707	0.5439	0.7450	1.8955
bc85856c012bcc1b	607_fri_c4_1000_50	667	50	10,209	pass	0.8856	0.9640	3.9394	—
c107ece39eb333a9d	584_fri_c4_500_25	333	25	5,638	pass	0.8062	0.8619	0.9417	—
c30ff9405c916718	579_fri_c0_250_5	167	5	1,692	pass	0.9475	1.0923	0.9961	0.9783
c4f8e81b70d33a13	590_fri_c0_1000_50	667	50	10,180	pass	0.8839	0.9172	1.0359	1.1762
c649a4cee87c4321	529_pollen	800	4	1,506	pass	1.0356	1.1012	0.9717	1.2459
cb5bf66ef2ee09ed	nikuradse_1	241	2	992	pass	0.9346	0.9373	0.9997	1.1909
cba8393879afd1dc	230_machine_cpu	140	6	2,338	pass	1.0026	1.0891	1.1718	—

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
ce1a79e2d658eb8a	strogatz_glider2	267	2	1,003	pass	0.6382	0.6625	1.0720	1.3756
ce837ccd24a3a18c	strogatz_bacres1	267	2	1,004	pass	0.7352	0.8309	0.9014	0.9826
cee6f811d0bf199d	feynman_II_11_3	800	5	1,728	pass	0.9385	0.9666	0.8950	1.0149
d255d3a2e1c8dbf4	feynman_III_12_43	800	2	1,028	pass	0.9920	0.9935	0.9966	0.9940
d52bfd2b6ab1c850	feynman_I_10_7	800	3	1,250	pass	0.9920	0.9932	0.9940	0.9959
d9a3408a840c9f48	560_bodyfat	168	14	4,364	pass	1.0247	1.0436	0.9930	1.3651
da290fae0d6c072f	feynman_test_10	800	3	1,267	pass	0.9955	0.9990	0.9990	1.0015
da78606950f100cd	1199_BNG_echoMonths	800	9	2,544	pass	0.9962	1.0086	0.9655	1.0694
dd16ac872ffb8673	strogatz_glider1	267	2	991	pass	0.1145	0.1723	0.9123	—
de2868ac837699fd	564_fried	800	10	2,861	pass	0.9477	0.9829	0.9543	1.4707
e08f0b9931f93686	225_puma8NH	800	8	2,439	pass	1.0639	1.1043	0.9951	2.8498
e18350fb51a9d1b0	strogatz_barmag2	267	2	1,027	pass	0.8897	0.9399	0.2855	—
e5a3a7b3668a1cc2	556_analcatdata_apnea2	316	3	1,238	pass	0.0339	0.0341	—	—
e68idd96cc63271b	573_cpu_act	800	21	4,954	pass	0.8080	0.9045	0.6593	1.1465
ed2efb82fce4b098	strogatz_vdp1	267	2	984	pass	0.5228	0.5812	1.2867	1.2401
edc9d41a298fd6f0	612_fri_c1_1000_5	667	5	1,705	pass	0.8224	0.8596	0.9670	1.2891
efd53e9c0f36d608	_deprecated_207_autoPrice	106	15	5,017	pass	0.9020	0.9357	1.0899	1.0398
f0a3d6881458f990	604_fri_c4_500_10	333	10	2,913	pass	0.6868	0.8360	0.8552	—
f24e9383331c43e2	503_wind	800	14	3,846	pass	0.9690	0.9962	1.0017	1.0367
f652ea96eff62654	strogatz_predprey1	267	2	989	pass	0.9964	1.1345	1.0218	1.4876
f8cf362e371e8960	strogatz_predprey2	267	2	1,030	pass	0.9876	0.9847	0.9718	1.0815
fc3812567b5fdd64	197_cpu_act	800	21	5,011	pass	0.7680	0.8922	0.6669	1.1253
fe6d7ac3fcd3129e	1201_BNG_breastTumor	800	9	2,427	pass	-0.6141	-0.3437	—	—

APPENDIX B

SUFFICIENCY BUDGET, ALL 100 LINEAGES

TABLE II: Sufficiency budget, all 100 lineages.

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
02a45777900441ce	215_2dplanes	800	10	—	—	—	—	—	—
047e62b52a3f4ede	695_chatfield_4	156	12	3,353	pass	1.0106	1.0287	1.0110	—
04c80944e971cb3e	first_principles_supernovae_zr	157	1	750	pass	0.9372	0.9454	1.0346	1.3271
0561d673cb73c27e	666_rmftsa_ladata	339	10	—	—	—	—	—	—
060b9ff8092db17	589_fri_c2_1000_25	667	25	5,734	pass	0.9237	0.9886	1.0258	1.5430
06b0312a585b436e	1196_BNG_pharynx	800	10	2,681	pass	0.8188	0.8768	1.0632	0.7968
0779d3fd5b3133a2	nikuradse_2	241	1	—	—	—	—	—	—
07f38946316c039b	593_fri_c1_1000_10	667	10	2,869	pass	0.9780	1.0222	0.9929	1.8097
0a8428ef25a69175	537_houses	800	8	2,636	pass	0.7243	0.7841	1.0073	1.0334
0d482c1dd7cc6592	210_cloud	72	5	1,855	pass	0.9561	0.8815	1.0056	0.9224
0db59fd2579600d0	665_sleuth_case2002	98	6	—	—	—	—	—	—
0f86a714ec5f3ac6	1191_BNG_pbc	800	18	4,218	pass	0.8069	0.8576	0.8783	—
167578f5deba4977	547_no2	333	7	2,474	pass	0.8821	0.9276	0.9850	1.0276
19f2c3c2d67defd7	feynman_II_11_27	800	4	—	—	—	—	—	—
19f4780b53b3fa11	678_visualizing_environmental	74	3	—	—	—	—	—	—
1a4456d1cc71a26a	feynman_test_11	800	7	2,185	pass	1.1162	1.3026	0.6721	—
1a803b27c2400dc4	feynman_III_10_19	800	4	1,522	pass	0.9698	0.9777	0.9732	0.9807
1cb97b1d7314d43d	229_pwLinear	133	10	2,598	pass	0.8275	1.0471	0.9727	—
205f4b4982b9ea49	608_fri_c3_1000_10	667	10	—	—	—	—	—	—
26eff843be35d831	581_fri_c3_500_25	333	25	5,655	pass	0.8660	0.9960	1.0900	1.6067
282f8ddb43b5e677	595_fri_c0_1000_10	667	10	2,913	pass	0.9645	1.0058	1.0029	1.1269
297b7688a045c583	603_fri_c0_250_50	167	50	10,183	pass	0.8257	1.1292	1.0240	2.8999
2c8d0985f3557d2b	feynman_II_11_28	800	2	—	—	—	—	—	—
30ca2887550f0d8d	201_pol	800	26	8,215	pass	0.6214	0.6436	0.7350	0.6603
3408a6a513c627a3	strogatz_barmag1	267	2	—	—	—	—	—	—
35a9a9b88f0e4d1	344_mv	800	10	—	—	—	—	—	—
3d3321e27d14f9aa	599_fri_c2_1000_5	667	5	1,713	pass	0.9820	1.0044	1.0441	1.6148
44eaabe4cb923766	613_fri_c3_250_5	167	5	1,708	pass	0.8605	0.9611	0.9427	—
46b25db9396c8209	227_cpu_small	800	12	3,338	pass	0.7033	0.8730	0.9142	0.9908
46c5a59e5ae42037	1193_BNG_lowbwt	800	9	2,468	pass	0.8393	0.8390	0.8798	0.8103
471f11cc2d38853a	feynman_test_11	800	4	1,472	pass	0.9340	0.9481	0.9730	0.9578
47c581af5de1136b	582_fri_c1_500_25	333	25	5,676	pass	0.8232	0.9820	0.8086	—
48b7a3bf81e5f3e5	605_fri_c2_250_5	167	25	5,696	pass	0.2608	0.5247	1.4268	—
4a354d98c965b01f	597_fri_c2_500_5	333	5	1,715	pass	0.9578	1.0001	1.0224	1.3200
4aa84fff8ad0cd26	feynman_III_13_18	800	4	—	—	—	—	—	—
51730b348b1bbd14	557_analcatdata_apnea1	316	3	1,246	pass	0.6556	0.6654	—	—
54b221668673164c	feynman_III_14_14	800	5	1,724	pass	1.0175	1.1226	1.0021	1.0727
552665868f8f72ed	strogatz_vdp2	267	2	—	—	—	—	—	—
585fef2a1cb6b33f	strogatz_lv2	267	2	1,021	pass	0.9239	1.0451	0.8041	—
642738a361f1c63a	feynman_I_12_1	800	2	1,034	pass	0.9936	0.9961	0.9993	0.9989
64c64f0f9983c48f	strogatz_bacres2	267	2	996	pass	0.9743	0.9826	0.9816	0.9923
654566a1f91408aa	4544_GeographicalOriginalofMusic	706	117	—	—	—	—	—	—
693890aae9dc860b	first_principles_supernovae_zg	162	1	766	pass	0.9415	0.9642	1.0277	1.1876
6cacb11611345a24	feynman_I_11_19	800	6	1,952	pass	0.9911	1.0176	1.0047	1.0089
7291b347fe0238fb	feynman_II_11_20	800	5	—	—	—	—	—	—
7423783f7e38bb9b	strogatz_shearflow1	267	2	1,023	pass	0.4860	0.5973	-4.3352	—
7a3d123dd555110d	598_fri_c0_1000_25	667	25	5,672	pass	0.8570	0.9206	1.0010	1.0168
7c8943b160464add	601_fri_c1_250_5	167	5	—	—	—	—	—	—
7e51711baa1b1fb1	feynman_I_12_11	800	5	1,720	pass	0.9780	0.9927	0.9910	1.0559
80589368fa78f21a	218_house_8L	800	8	2,410	pass	0.6661	0.7528	0.6830	1.0728

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
8164950b155f7f56	feynman_I_12_2	800	4	—	—	—	—	—	—
85a31ae409944a1b	feynman_II_10_9	800	3	1,264	pass	0.9803	0.9961	0.9881	1.0061
8d859cfa7b51a351	feynman_III_15_12	800	3	1,280	pass	0.7428	0.8572	0.8771	1.2306
923483635406b213	1203_BNG_pwLinear	800	10	—	—	—	—	—	—
9312d813229df035	690_visualizing_galaxy	216	4	1,470	pass	0.9770	0.9875	0.9843	1.0661
94a00ea67f49bf84	596_fri_c2_250_5	167	5	1,718	pass	0.8262	0.8934	—	—
94bd4520b8ceb3f0	feynman_test_12	800	5	1,745	pass	0.9677	0.9845	0.9937	1.0157
956d34f13e7af830	586_fri_c3_1000_25	667	25	—	—	—	—	—	—
9a10c15885022034	_deprecated_195_auto_price	106	15	5,038	pass	0.8965	0.9544	0.9901	1.3165
a04f964bc53281e0	strogatz_lv1	267	2	986	pass	-3.0531	-2.4502	—	—
a23f4bd626b76590	feynman_test_13	800	5	—	—	—	—	—	—
a32a69e7c701eb14	588_fri_c4_1000_100	667	100	19,041	fail	—	—	—	—
a5ba682572f53add	602_fri_c3_250_10	167	10	2,902	pass	0.7774	0.9773	0.9471	—
a863cd42be4d05d4	663_rabe_266	80	2	1,014	pass	0.9849	0.9996	1.0018	1.0573
ab7065df07a47eb0	592_fri_c4_1000_25	667	25	—	—	—	—	—	—
ab279eded3b3c52b	574_house_16H	800	16	4,076	pass	0.0805	0.1211	-0.7472	-0.3199
aebf5d861a8365b4	522_pm10	333	7	2,450	pass	0.6372	0.7959	0.6203	3.4124
b6adf79118a58cea	562_cpu_small	800	12	—	—	—	—	—	—
b78d323566930bd7	583_fri_c1_1000_50	667	50	10,098	pass	0.8817	0.9737	0.8733	0.9465
b7f9774e2142a758	strogatz_shearflow2	267	2	1,022	pass	0.7388	0.8339	-0.4873	1.5681
bc85856c012bce1b	607_fri_c4_1000_50	667	50	—	—	—	—	—	—
c107ee39eb333a9d	584_fri_c4_500_25	333	25	5,641	pass	0.8307	0.9024	0.9623	—
c30ff9405c916718	579_fri_c0_250_5	167	5	1,700	pass	0.9081	1.0750	0.9949	0.9830
c4f8e81b70d33a13	590_fri_c0_1000_50	667	50	—	—	—	—	—	—
c649a4cee87c4321	529_pollen	800	4	1,506	pass	1.0619	1.1198	0.9983	1.2801
cb5bf66ef2ee09ed	nikuradse_1	241	2	990	pass	0.9662	0.9677	0.9970	1.1953
cba8393879afd1dc	230_machine_cpu	140	6	—	—	—	—	—	—
ce1a79e2d658eb8a	strogatz_glider2	267	2	1,004	pass	0.5396	0.6099	1.1474	1.2017
ce837ccd24a3a18c	strogatz_bacres1	267	2	1,004	pass	0.7807	0.8568	0.8858	0.9570
cee6f811d0bf199d	feynman_II_11_3	800	5	—	—	—	—	—	—
d255d3a2e1c8dbf4	feynman_III_12_43	800	2	1,023	pass	0.9947	0.9964	0.9984	0.9967
d52bfd2b6ab1c850	feynman_I_10_7	800	3	1,248	pass	0.9978	0.9988	0.9994	1.0005
d9a3408a840c9f48	560_bodyfat	168	14	—	—	—	—	—	—
da290fae0d6c072f	feynman_test_10	800	3	1,257	pass	0.9681	0.9718	0.9737	0.9744
da78606950f100cd	1199_BNG_echoMonths	800	9	2,578	pass	0.9473	0.9673	0.9447	1.0028
dd16ac872ffb8673	strogatz_glider1	267	2	—	—	—	—	—	—
de2868ac837699fd	564_fried	800	10	2,864	pass	0.9826	1.0201	0.9548	1.5094
e08f0b9931f93686	225_puma8NH	800	8	2,440	pass	1.0168	1.0664	0.9433	2.7529
e18350fb51a9d1b0	strogatz_barmag2	267	2	—	—	—	—	—	—
e5a3a7b3668a1cc2	556_analcatdata_apnea2	316	3	1,241	pass	0.6282	0.6707	—	—
e68fdd96cc63271b	573_cpu_act	800	21	4,958	pass	0.8507	0.9689	0.5661	1.1177
ed2efb82fce4b098	strogatz_vdp1	267	2	991	pass	0.5397	0.6472	-0.1705	1.1154
edc9d41a298fd60	612_fri_c1_1000_5	667	5	—	—	—	—	—	—
efd53e9c0f36d608	_deprecated_207_autoPrice	106	15	5,024	pass	0.9035	0.9215	1.0781	1.0481
f0a3d6881458f990	604_fri_c4_500_10	333	10	2,918	pass	0.8895	1.0064	1.0421	—
f24e9383331c43e2	503_wind	800	14	3,877	pass	0.9992	0.9953	0.9918	1.0034
f652ea96eff62654	strogatz_predprey1	267	2	—	—	—	—	—	—
f8cf362e371e8960	strogatz_predprey2	267	2	1,030	pass	0.9865	0.9827	0.9658	1.0750
fc3812567b5fdd64	197_cpu_act	800	21	5,006	pass	0.8164	0.9529	0.6632	1.1726
fe6d7ac3fcd3129e	1201_BNG_breastTumor	800	9	—	—	—	—	—	—

APPENDIX C

BEYONDARENA FAMILIES IN THE DATASET BUCKET

TABLE III: All 142 BeyondArena families.

name	task	rows	cols	license	split
5g_energy_consumption	regression	92,629	22	MIT	grouped
acquire_valued_shoppers_challenge	binary	160,057	112	not cleared	temporal
airfoil_self_noise	regression	1,503	6	CC-BY-4.0	iid
allstate_claims_severity	regression	188,317	131	not cleared	iid
amazon_employee_access	binary	32,769	10	public-domain	iid
amex_non_iid	binary	1,249,605	191	not cleared	grouped
anes_voting_2026	binary	48,587	319	not cleared	temporal
aps_failure	binary	76,000	171	CC-BY-4.0	iid
asp_potassco_classification	multiclass	1,212	138	not cleared	grouped
audiology_diagnosis	multiclass	199	69	CC-BY-4.0	iid
bad_customer_detection	binary	1,723	14	public-domain	iid
bank_customer_churn	binary	10,000	11	public-domain	iid
bank_marketing	binary	45,211	14	CC-BY-4.0	iid
biogeographical_ancestry_prediction	multiclass	635	105	not cleared	iid
biomechanical_orthopaedic_prediction	multiclass	310	7	CC-BY-4.0	iid
bioresponse	binary	3,751	1777	public-domain	iid
blood_tests_drink_prediction	regression	345	6	CC-BY-4.0	iid
blood_transfusion	binary	748	5	CC-BY-4.0	iid
body_density_prediction	regression	252	14	not cleared	iid
california_house_prices_2020	regression	41,528	42	not cleared	temporal
cardiotocography	multiclass	2,126	24	CC-BY-4.0	grouped
churn	binary	5,000	20	public-domain	iid
cirrhosis_patient_survival_prediction	regression	161	18	not cleared	iid

TABLE III, continued

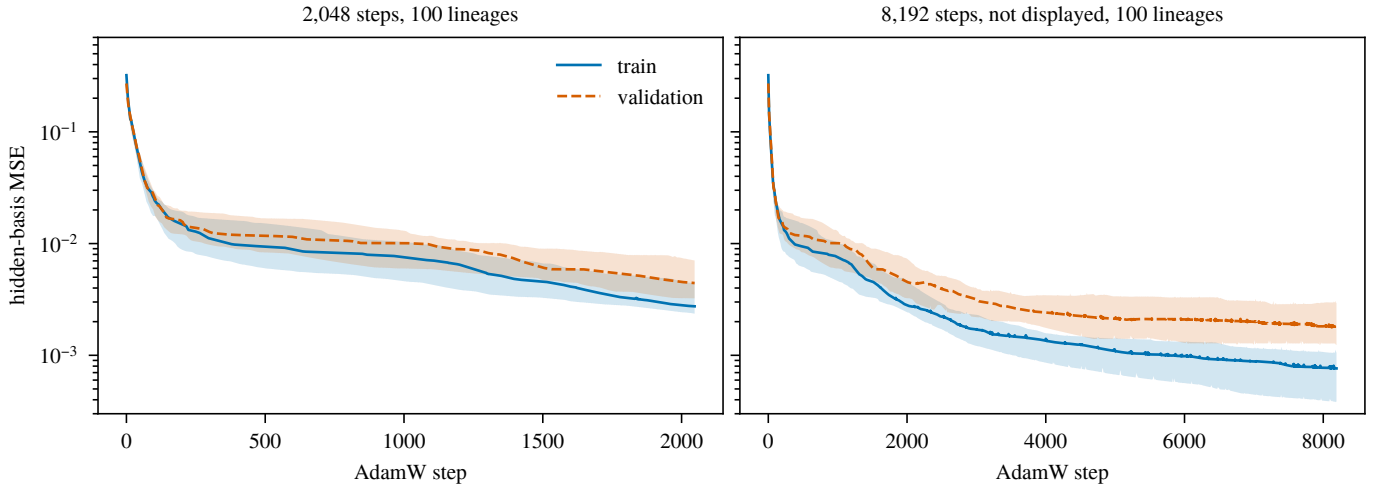
name	task	rows	cols	license	split
climate_model_weather_forecasting	regression	1,250,000	101	not cleared	temporal
clock_protein_toxicity	binary	171	1118	CC-BY-4.0	iid
coffee_rating_prediction	regression	2,369	13	not cleared	temporal
coil_2000	binary	9,822	86	CC-BY-4.0	iid
concrete_compressive_strength	regression	1,030	9	CC-BY-4.0	iid
consumer_complaints	multiclass	1,226,140	13	us-gov	temporal
cooking_time	regression	1,250,000	197	not cleared	temporal
covertime	multiclass	512,625	15	CC-BY-4.0	grouped
credit_approval	binary	690	16	CC-BY-4.0	iid
credit_card_clients_default	binary	30,000	24	CC-BY-4.0	iid
credit_g	binary	1,000	21	CC-BY-4.0	iid
customer_satisfaction_in_airline	binary	129,880	22	public-domain	iid
delivery_eta	regression	1,250,000	226	not cleared	temporal
dementia_prediction	multiclass	370	10	not cleared	grouped
diabetes_130_us	binary	71,518	45	CC-BY-4.0	iid
diamonds	regression	53,940	10	MIT	iid
drug_induced_autoimmunity_prediction	binary	597	178	CC-BY-4.0	iid
early_learning_predictors	regression	18,874	745	CC-BY-4.0	grouped
early_stage_diabetes_risk_prediction	binary	251	17	CC-BY-4.0	iid
ecoli_proteins	multiclass	327	7	CC-BY-4.0	iid
ecommerce_shipping	binary	10,999	11	public-domain	iid
electric_motor_temperature_prediction	regression	1,296,316	111	not cleared	grouped
emscad	binary	17,460	18	CC0-1.0	iid
eryhemato_squamous_disease	multiclass	366	35	CC-BY-4.0	iid
fiat_500	regression	1,538	8	CC0-1.0	iid
fitness_club	binary	1,500	7	public-domain	iid
food_delivery_time	regression	45,451	10	not cleared	iid
forensic_glass_identification	multiclass	214	10	CC-BY-4.0	iid
forest_fires	regression	517	13	CC-BY-4.0	iid
gallstone_disease	binary	319	39	CC-BY-4.0	iid
garments_worker_productivity	regression	1,197	16	CC-BY-4.0	temporal
ghanas_indigenous_intel	multiclass	10,928	11	not cleared	temporal
give_me_some_credit	binary	150,000	11	not cleared	iid
hazelnut_spread_contaminant_detection	binary	2,400	31	not cleared	iid
healthcare_insurance_expenses	regression	1,338	7	not cleared	iid
heart_disease_cleveland	binary	303	14	CC-BY-4.0	iid
heart_disease_hungary	binary	294	14	CC-BY-4.0	iid
heart_disease_va_long_beach	binary	200	14	CC-BY-4.0	iid
heart_failure_followup_survival	binary	299	13	CC-BY-4.0	iid
heloc	binary	10,459	24	not cleared	iid
hepatitis_c_prediction	multiclass	608	13	CC-BY-4.0	iid
hepatitis_survival_prediction	binary	155	20	CC-BY-4.0	iid
hiva_agnostic	binary	3,845	1519	public-domain	iid
home_credit_default_risk	binary	307,507	505	not cleared	iid
home_credit_default_stability	binary	1,224,927	712	not cleared	temporal
homeq_default_prediction	binary	5,708	13	not cleared	iid
homesite_quote_conversion	binary	260,753	296	not cleared	iid
horse_colic_survival	multiclass	344	21	CC-BY-4.0	iid
hotel_booking_demand	binary	81,418	32	CC-BY-4.0	temporal
houses	regression	19,675	9	not cleared	iid
hr_analytics	binary	19,158	13	CC0-1.0	iid
ieee_fraud_detection	binary	590,540	436	not cleared	temporal
immoscout_german_house_prices	regression	10,317	24	not cleared	iid
in_vehicle_coupon_recommendation	binary	12,684	25	CC-BY-4.0	iid
indian_liver_patient_dataset	binary	583	11	CC-BY-4.0	iid
iranian_churn	binary	2,850	14	CC-BY-4.0	iid
jml	binary	10,885	22	public-domain	iid
kdd_cup_09_appetency	binary	50,000	213	public-domain	iid
kick	binary	72,983	33	public-domain	temporal
kickstarter	binary	187,118	16	not cleared	temporal
labour_inspection_compliance	binary	63,634	377	CC0-1.0	iid
lending_club	binary	1,064,751	97	CC0-1.0	temporal
ljubljana_breast_cancer	binary	286	10	CC-BY-4.0	iid
ljubljana_primary_tumor	multiclass	302	18	CC-BY-4.0	iid
lung_cancer	multiclass	197	12601	not cleared	iid
lung_cancer_epithelial_genexp	binary	187	22216	public-domain	iid
maps_router_eta	regression	1,250,000	989	not cleared	temporal
marketing_campaign	binary	2,240	26	CC-BY-4.0	iid
maternal_health_risk	multiclass	1,014	7	CC-BY-4.0	iid
mercari_price_suggestion	regression	1,250,000	7	not cleared	iid
mercedes_benz_greener_manufacturing	regression	4,204	372	not cleared	temporal
miami_housing	regression	13,776	16	not cleared	iid
mic	multiclass	1,699	112	CC-BY-4.0	iid
mice_protein_trisomy_discriminant	multiclass	1,080	78	CC-BY-4.0	grouped
micro_mass	multiclass	571	1084	CC-BY-4.0	grouped
musk	binary	6,598	168	CC-BY-4.0	grouped
mutual_funds_india	regression	793	13	CC0-1.0	iid
naticusdroid_android_permissions_dataset	binary	7,491	86	CC-BY-4.0	iid
obesity_estimation	regression	498	15	CC-BY-4.0	iid
online_shoppers_purchasing_intention_dataset	binary	12,330	18	CC-BY-4.0	iid
otto_group_product_classification_challenge	multiclass	61,878	94	not cleared	iid
pancreatic_cancer_mouse_detection	binary	181	6773	not cleared	grouped
parkinsons_biomedical_voice_measurements	binary	195	25	CC-BY-4.0	grouped

TABLE III, continued

name	task	rows	cols	license	split
physiochemical_protein	regression	45,730	10	CC-BY-4.0	iid
polish_companies_bankruptcy	binary	5,790	65	CC-BY-4.0	iid
porto_seguro	binary	595,206	38	not cleared	iid
predict_students_dropout_and_academic_success	multiclass	4,424	37	CC-BY-4.0	iid
prostate_cancer_detection	binary	322	15155	not cleared	iid
pva_revenue_prediction_kddcup98	binary	144,095	478	not cleared	iid
qsar_aquatic_toxicity	regression	546	9	CC-BY-4.0	iid
qsar_biodeg	binary	1,054	42	CC-BY-4.0	iid
qsar_fish_toxicity	regression	908	7	CC-BY-4.0	iid
qsar_tid_11	regression	5,741	1025	public-domain	iid
regensburg_pediatric_appendicitis	binary	763	52	not cleared	iid
rossmann_store_sales	regression	844,392	16	not cleared	temporal
santander_customer_satisfaction	binary	71,080	308	not cleared	iid
santander_customer_transaction_prediction	binary	200,000	601	not cleared	iid
santander_transaction_value	regression	4,447	541	not cleared	iid
sat11_hand_algo_runtime	regression	2,960	171	not cleared	grouped
sberbank_housing_market_forecasting	regression	27,195	387	not cleared	temporal
sdss_17	multiclass	78,053	12	public-domain	iid
seismic_bumps	binary	2,584	16	CC-BY-4.0	iid
sepsis_prediction	binary	1,228,686	44	not cleared	grouped
sepsis_survival_minimal_clinical_records	binary	110,204	4	CC-BY-4.0	iid
sf_permit_time	regression	116,954	38	not cleared	temporal
south_africa_coronary_heart_disease	binary	462	10	not cleared	iid
splice	multiclass	3,190	61	CC-BY-4.0	iid
student_portuguese_performance	regression	649	31	CC-BY-4.0	iid
superconductivity	regression	21,263	82	CC-BY-4.0	iid
taiwanese_bankruptcy_prediction	binary	6,819	93	CC-BY-4.0	iid
telemonitoring_parkinsons_biomedical_voice_measurements	regression	502	21	CC-BY-4.0	grouped
thyroid_discordant	binary	3,711	27	CC-BY-4.0	iid
tour_travels_churn	binary	954	7	CC0-1.0	iid
video_game_fps_prediction	regression	12,288	40	CC-BY-4.0	grouped
video_transcoding_time_prediction	regression	68,784	20	CC-BY-4.0	grouped
website_phishing	multiclass	1,353	10	CC-BY-4.0	iid
wids_diabetes_mellitus	binary	127,358	182	not cleared	iid
wine_quality	regression	6,497	13	CC-BY-4.0	iid
wine_world_cost	regression	1,279	15	CC0-1.0	iid

APPENDIX D REPLAY CURVES

The journal reports the median hidden-basis loss. The curves below are the logged replay of both step budgets, before the readout refit. The 8,192-step panel is not the displayed model. The band is a 95% lineage-bootstrap interval. All 100 lineages finished. Median training loss starts at 0.321 and ends at 0.00274 and 0.000763. Median validation loss starts at 0.271 and ends at 0.00443 and 0.00181. Where both budgets have a complete three-seed group, the linear medians are 0.990 and 0.987 (64 lineages; difference -0.000283). The MLP medians are 1.065 and 1.057 (52 lineages; difference 0.00451).



APPENDIX E REPRODUCIBILITY

Regenerate every generated/ file, the figure PDFs, and both manuscript PDFs with `just paper` from the commit that contains these files. That recipe reads the committed validation ledgers. When the benchmark scratch is mounted it also rereads the fit receipts and the pinned LICENSE files. The PDF build reads the committed extracts and does not invent a missing timer or license. It does not open an official test file. The catalog hash is `ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100`

df34377e81aa1120. The grouped training split uses seed 1729. The measured fit seed is 11, and the sample seeds are 101, 211, and 307. `python3 research/benchmark/verify_paper_numbers.py` exits nonzero when a displayed macro disagrees with those ledgers. Placeholder cells for TabSyn, a full TabDDPM panel, the production privacy-attack panel, fit-seed variance, and the full ablation grid stay the words “not measured” until those runs exist. A receipt field that was not stored is “not recorded.” The journal fidelity table is a separate empirical ledger on the grouped validation split. It is not the production privacy-attack panel, and it is not differential privacy or HIPAA de-identification.

PMLB is MIT. Every dataset entry in the data lock records SPDX MIT and the note that the source license is MIT (PMLB). CTGAN and TVAE share one pinned LICENSE file: Business Source License 1.1, licensor DataCebo, Inc., licensed work CTGAN. The Gaussian copula uses the pinned Copulas LICENSE: Business Source License 1.1, licensor DataCebo, Inc., licensed work Copulas. TabSyn is Apache License 2.0. TabDDPM is MIT License, Copyright (c) 2022 Akim Kotelnikov. Forest-Diffusion is MIT License, Copyright (c) 2023 Samsung Electronics Co., Ltd. All Rights Reserved. ARF is MIT License, Copyright (c) 2023 Kristin Blesch, Marvin Wright. Chow-Liu and the independent marginals are study code under this repository’s license: MIT License, Copyright (c) 2026 Dope contributors. The SDV repository LICENSE is not in the pinned snapshot, so that license is not recorded.

Peak RAM in the compute table is resident bytes, shown in mebibytes to one decimal, and peak VRAM is `peak_gpu_used_mib`. The hardware cell drops a leading NVIDIA GeForce prefix. GPU hours sum walls on which the receipt recorded a GPU process, and each DOPE fit file is counted once. CPU hours stay not recorded where the receipt has no separate CPU timer. The logged replay medians 4.56 and 14.4 are not the scored campaign (not measured). Forest-Flow medians use the 12 native-grid operation walls. Its bytes are the 6 selected artifacts. 953 sums `fit_seconds` and 8,801 sums receipt operations. Neither sum is that median. Whole-device energy is not job energy. Gaussian copula, Chow-Liu, and independent-marginal attempts store one unsplit `wall_seconds`. Medians are 1.71, 0.168, and 0.162. Maxima are 39.9, 1.29, and 0.429. BeyondArena medians remain 5.00 and 14.0. Reading the journal threshold counts from top to bottom, CatBoost is 97, 97, 97; linear is 93, 92, 90; and the MLP is 75, 75, 73.

APPENDIX F FIT CORESETS

The fit view is not the full PMLB table. Frozen catalog metadata draws each official training and test file by quota stratification on the target distribution and high-variance feature quantile bins, aiming at 1000 training rows and 1000 test rows, with at most 2000 rows in total and at least 101 rows. A source smaller than that quota keeps every row and still reserves a test share of at least 0.2. Of the 100 prepared lineages, 60 were already inside the quota. The other lineages use a hybrid coreset that keeps the observed feature and target range; the bin counts are 1 lineage at 11 bins, and 39 lineages at 50 bins. The generator then groups identical official-training rows and assigns those groups with seed 1729 so that about 80% of the rows become the fit view and the rest become the validation holdout. Duplicate rows are not split across that cut. On these lineages the fit view has 72 to 800 rows. The bound keeps every lineage on one worker view and leaves the official test sealed. It is not a second draw at scoring time.

APPENDIX G ARTIFACT HASHES

The rows below are SHA-256 of each ledger and round lock the paper reads, plus the commit and the file hash of the harness that measured the grouped-validation fidelity ledger. Regenerating the macros does not rewrite a ledger hash.

Ledger or lock	SHA-256
Code commit of the measured harness	17f3294d139e19f0ec831605f8750368eebc30bb
Harness SHA-256	a952062c0f83805f6442a440a5ae15293a843349d423eaa106a9c56a20d14c0f
<code>research/benchmark/expanded_validation_metrics.py</code>	a952062c0f83805f6442a440a5ae15293a843349d423eaa106a9c56a20d14c0f
<code>research/benchmark/methods.lock.json</code>	899c355cb55769710ed369fea6b082589b31858860fb74929bf4190de28aca33
<code>research/benchmark/results/expanded-validation-diagnostics.json</code>	cd20ac0d07a1dcab7c5e46db195adb463172b5174387bb71a054b2e5f340beeb
<code>research/benchmark/results/s3-coreset-procedure.json</code>	73251e43cf7838687427569931bf6127f42e932374af33137c57fd8fe8544704
<code>research/benchmark/results/s3-lineage-record.json</code>	bbd0852c49f595315104261efa4eaa6f50db257e323b8c002aac6737a2dbe7d7
<code>research/benchmark/results/density-matched-population-validation.json</code>	5264b88a40ad5efb21d9b119789caeb13e71e911a17f1d8b57f1d8e8ac31732a
<code>research/benchmark/results/sdv-matched-population-validation.json</code>	4dc367eb78159a0386882d23ae1f99b0aa2cc9f425d1aae499b6f3453222a44e
<code>research/benchmark/results/s3-matched-forest-confirmation-validation.json</code>	97a25be902954cad16c5b5802d0e0ba468dea104221efa2465434c94e6774a1f
<code>research/benchmark/results/s3-data.lock.json</code>	857b61324d0a5b7dbbb98e0c147fcc79c5bb8616e7ca61afd6785469c17f038f
<code>research/benchmark/results/arf-native-closure-watch-v1.receipt.json</code>	7e8eeb1ba94ca747367e98d5a016d2ed3ed4c40c5a98ca76d7a5fe27ce9c173c

Ledger or lock	SHA-256
research/benchmark/results/beyondarena-s3-inventory.json	504166764631dbdf5e02d116334af5cc587731fcf3efe130d62240e6dd512d64
docs/whitepaper/generated/fit-trace.json	d8c31b9e4eb59cb1a087cf5dcdde4a91a73eca00f3704fbff647a7da6ce65522
docs/whitepaper/generated/loss-curves.json	0124915497c98e560ef99c1e5ecf7ad876f9174439ff9c763cf1cf906b4aebel
docs/whitepaper/generated/replay-cost.json	faf77e7celed1d22127701bf68bbfe2f8d183f89cbd3475dd80094888c36f086
docs/whitepaper/generated/provenance.json	6880d7d3595c0ab9ada7396db05513bc4be538278e44a510334e50e41c7dc7f7
docs/whitepaper/generated/beyond-fit.json	a1447f908bc9c51876d345597039607c734b507bf85aa8e08db0639e80a1f036